

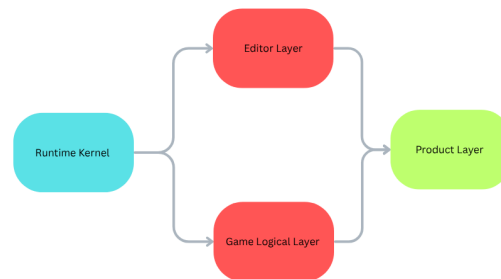
# Omnix Engine

V0.6

Architectural Upgrades

# Architectural Upgrade

Omnix Engine is now moving to a drastic architectural upgrade . From we are using the given architectural design for the whole Omnix Project .



So the whole engine is divided into 3 layers :

1. Runtime Kernel Layer
2. Editor and Game Logic Layer [ Development Layer ]
3. Product Layer

# Runtime Kernel Layer

As the name suggests , it is the core layer which contains the key functionalities to power and make any engine or game using this framework .

So this is the architecture of OmnixKernelEngine [ OKE ]

## Omnix Runtime Kernel

- |
- |—— 1. Core Foundation
- |—— 2. Platform Layer
- |—— 3. Application Framework
- |—— 4. Asset & Resource System
- |—— 5. Scene & ECS
- |—— 6. Rendering
- |—— 7. Physics
- |—— 8. Animation
- |—— 9. Audio
- |—— 10. Input
- |—— 11. AI Foundation
- |—— 12. Networking Foundation
- |—— 13. Runtime Services
- |—— 14. Developer Services
- |—— 15. Packaging & Deployment
- |—— 16. Plugin & Extension Framework

## Core Foundation

This should be rock solid because everything else depends on it.

Memory Management

Containers

Math Library

UUID System  
Reflection  
Serialization  
Logging  
Assertions  
Error Handling  
Event System  
Time System  
Threading  
Job System  
Task Graph  
Configuration System  
Command Line Parser  
File Utilities

## Platform Layer

Window Management  
Monitor Enumeration  
Display Modes  
Input Backend  
File System  
Dynamic Library Loader  
Clipboard  
Cursor  
Timers  
Threads  
Environment Variables  
Platform Detection

# Application Framework

Engine  
Application  
Layer Stack  
Module Manager  
Startup  
Shutdown  
Main Loop  
Update Phases  
Fixed Update  
Render Update  
Tick System

## Scene and ECS

Scene Graph  
Entity Manager  
Component Manager  
Systems Manager  
World  
Prefabs  
Scene Serialization  
Scene Streaming  
Entity Hierarchy  
Runtime Instantiation  
Runtime Destruction

## Rendering

RHI  
Swapchain  
Command Buffers  
Synchronization

Render Graph  
GPU Scene  
Materials  
Shaders  
Pipeline Cache  
Textures  
Meshes  
Virtual Geometry  
Visibility  
Lighting  
Shadows  
Post Processing  
Debug Rendering  
GPU Profiling

## Physics

- Physics World
- Rigid Bodies
- Character Controller
- Shapes
- Collision Detection
- Triggers
- Raycasts
- Sweep Tests
- Constraints
- Physics Materials
- Debug Rendering

## Animation

- Skeletons
- Bones

- Animation Clips
- Animation Player
- Blend Trees
- State Machines
- Animation Graph
- Root Motion
- Animation Events
- GPU Skinning
- Morph Targets
- IK Foundation

## Audio

Audio Device

Audio Sources

Audio Listener

Audio Mixer

Streaming Audio

3D Spatial Audio

Reverb Zones

Occlusion

DSP Effects

Audio Debug

## Input

Keyboard

Mouse

Gamepad

Input Mapping

Input Contexts

Actions

Axes  
Hot Plugging  
Haptics

## AI Foundation

Navigation Mesh  
Pathfinding  
Blackboard  
Behavior Tree Runtime  
EQS Runtime (optional)  
Crowd Navigation  
Steering Behaviors

## Networking Foundation

Socket Layer  
Replication Framework  
RPC Support  
Serialization  
Network Time  
Prediction Hooks  
Client/Server Framework

## Runtime Services

Save System  
Configuration  
Localization  
Plugin Manager  
Module Loader  
Crash Handler  
Runtime Console  
Runtime Statistics



Version System  
Feature Flags

## Developer Services

Profiler  
Memory Profiler  
GPU Profiler  
CPU Profiler  
Debug Renderer  
Statistics  
Asset Validator  
Performance Markers  
Diagnostic Logging

## Packaging and Deployment

Package Format  
Package Reader  
Compression  
Encryption (optional)  
Version Compatibility  
Runtime Mounting  
Dependency Validation

# Development Layer

## 1. Project Management

The entry point of Omnix.

### Features

- Project Browser
- Create Project Wizard
- Open Existing Project
- Recent Projects
- Project Templates
- Plugin Management
- Engine Version Manager
- Project Migration
- Project Validation
- Build Configuration
- Workspace Settings

Comparable to: Unreal Project Browser

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## 2. World Editor

The primary authoring environment.

### Features

- Scene Viewport
- World Outliner
- Entity Hierarchy
- Entity Inspector
- Gizmos

- Grid
- Snapping
- Layers
- Groups
- World Settings
- Multi-selection
- Search

Comparable to: Unreal Level Editor

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## 3. Asset Management

Everything related to content.

Features

- Asset Browser
  - Asset Database
  - Asset Preview
  - Asset Metadata
  - Dependency Viewer
  - Reference Viewer
  - Asset Search
  - Favorites
  - Collections
  - Redirectors
  - Asset Validation
  - Duplicate Finder
- 

## 4. Import & Cooking Pipeline

Transforms source assets into runtime assets.

## Features

- Mesh Import
  - Texture Import
  - Material Import
  - Animation Import
  - Audio Import
  - Batch Import
  - Reimport
  - Asset Cooking
  - Compression
  - Optimization
  - Platform Conversion
- 

## 5. Visual Editors

One of the largest parts of the Development Layer.

### Material Editor

- Node Graph
  - Preview
  - Parameters
  - Instances
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### Animation Editor

- Skeleton Viewer
- Animation Timeline
- Animation Preview
- Blend Trees
- State Machines

- Animation Events
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## Shader Editor

- Graph
  - Live Compilation
  - Error View
- 

## Particle Editor

- Emitters
  - Modules
  - Curves
- 

## UI Editor

- Canvas
  - Widgets
  - Anchors
  - Styling
- 

## Audio Editor

- Mixer
  - Events
  - Buses
- 

## Terrain Editor

- Sculpting
  - Painting
  - Layers
- 

## 6. Gameplay Authoring

These are development tools, not runtime systems.

### Features

- Prefab Editor
  - Entity Templates
  - Gameplay Tags Editor
  - Data Table Editor
  - Dialogue Editor
  - Quest Editor
  - Localization Editor
  - Input Mapping Editor
  - Save Data Inspector
- 

## 7. Cinematics

### Features

- Timeline
- Sequencer
- Camera Tracks
- Animation Tracks
- Audio Tracks
- Events
- Keyframes
- Cutscene Preview

Comparable to Unreal Sequencer.

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## 8. AI Authoring

Editor tools for runtime AI.

Features

- NavMesh Bake
  - NavMesh Preview
  - Behavior Tree Editor
  - Blackboard Editor
  - EQS Editor
  - AI Debugger
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## 9. Debugging & Profiling

A complete development suite.

Features

- Console
  - Statistics
  - GPU Profiler
  - CPU Profiler
  - Memory Profiler
  - Asset Profiler
  - Frame Debugger
  - RenderDoc Integration
  - Performance Timeline
  - Debug Draw
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## 10. Build & Deployment

Everything needed to produce a shipping build.

### Features

- Build Profiles
  - Packaging
  - Cooking
  - Platform Targets
  - Versioning
  - Installer Generation
  - Release Notes
  - Symbol Generation
  - Build Reports
- 

## 11. Automation

Large studios depend heavily on automation.

### Features

- Automation Tests
  - Unit Tests
  - Performance Tests
  - Regression Tests
  - Commandlets
  - Batch Processing
  - Build Automation
  - Asset Validation Automation
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## 12. Developer Experience



Quality-of-life tools.

## Features

- Docking
- Themes
- Shortcuts
- Workspace Layouts
- Plugin Browser
- Documentation Browser
- Search Everywhere
- Command Palette
- Notifications
- Live Logs